

Power Electronics Laboratory

Session 2: DC-DC converter 2

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Objectives

1. Study the impact of losses and disturbances on the operation of the power converter.
2. Introduce the internal inductor current controller
3. Introduce the external capacitor voltage controller.
4. Assess the benefits of the controllers on the operation of the power converter under disturbances.

General instructions

1. Introduction is evaluated with **0.5** points maximum.
2. Conclusion is evaluated with **0.5** points maximum.
3. For every **3** figure format and order error, **0.1** points will be discounted.
4. All figures and equations must be referenced.
5. In many tasks, it is not indicated what variables must be plotted. The student must be able to decide what variables to plot, and how to plot them.

Part 1

Build the circuit shown in Fig. 1, using the parameters of Table 1. To control the switches of the power converter, implement one of the PWMs presented in the Session 1.

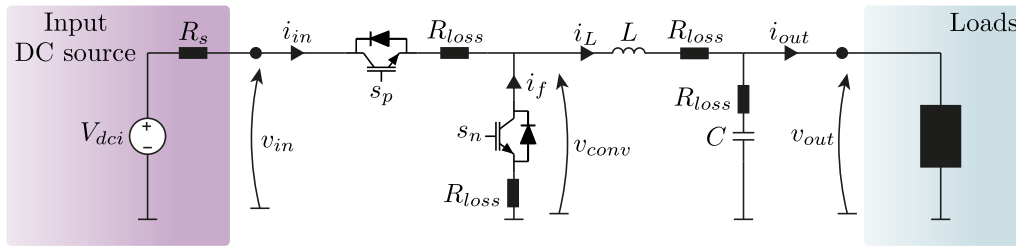


Figure 1: Nonideal Synchronous Buck converter.

Table 1: System parameters.

Parameter	Value
Input voltage	500V
Input source resistance	0.1
Inductance	200 μ H
Capacitance	100 μ F
Equivalent losses resistance	50m Ω
Switching frequency	10KHz
Resistive load	5 Ω

Tasks

1. Choose a reference duty cycle, d^* , and show the transient response of the system to:
 - Load step change.
 - DC input step voltage disturbance.

In the results, show the system state before and after the disturbances. Therefore, configure your scope properly. Additionally, decide and comment on what signals will be measured. Discuss your results.

Part 2

In **Part 1** it was shown that disturbances such as load steps and input voltage variations affect the performance of the power converter. Specifically, the output voltage is degraded, deviating from its expected ideal value. In many applications, the DC voltage must be regulated and kept as constant as possible at a desired value, independently of disturbances.

In the literature, it is possible to find different methods to control the output voltage of the power converter. In this laboratory, a cascaded configuration, shown in Fig. 2, will be implemented.

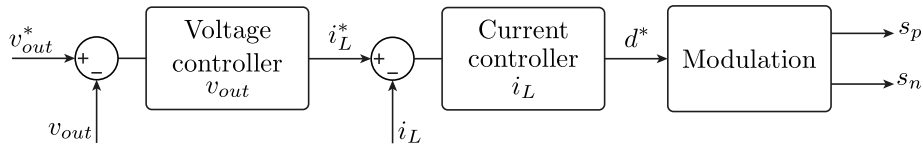


Figure 2: Cascaded voltage and current control diagram.

The controller can be summarized as follows:

- An internal controller regulates the inductor current of the power converter, i_L , by modifying the reference duty cycle, d^* . This controller is designed to follow a current reference, i_L^* , without error in steady state, and it is typically designed in order to obtain a fast response to current reference step changes, in the order of milliseconds.
- The external controller regulates the output voltage (capacitor voltage). This is achieved by dynamically modifying the inductor current reference. In steady state, the controller must be able to follow, without error, the output voltage reference, v_{out}^* . A correct and stable performance is achieved when the external controller is at least 10 times slower than the internal controller, i.e., it should be designed to achieve the steady state in the order of tenths of milliseconds.

In this part, the internal inductor current controller will be implemented. Therefore, the circuit must be slightly modified. Implement the circuit shown in Fig. 3. Observe that the output capacitor and load were replaced by an ideal voltage source.

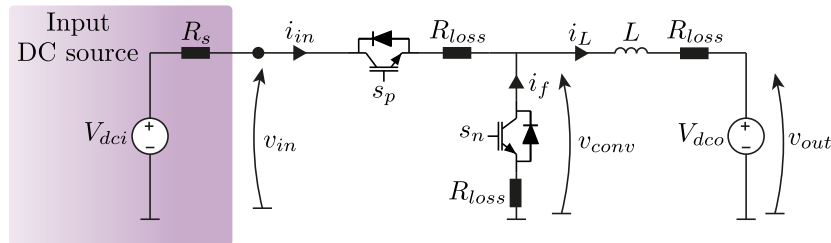


Figure 3: Modified buck for current control testing.

In this laboratory, the internal controllers are based on a Proportional Integral (PI) controller with output voltage feedforward. Implement the control diagram shown in Fig. 4 with the parameters shown in Table 2.

Observe in Fig. 4 that the output of the control stage corresponds to the reference converter voltage, v_{conv}^* . This is the voltage that will be modulated by the PWM stage. Therefore, this voltage must be scaled down in order to fit in the range of the carrier of the PWM generator. The down-scaling is done

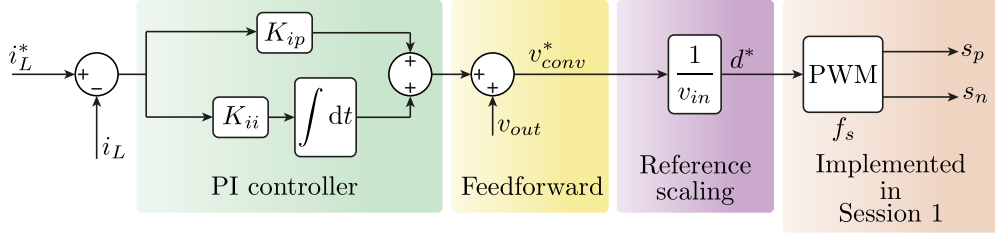


Figure 4: Internal inductor current controller with output voltage feedforward.

Table 2: Current controller parameters

Parameter	Value
Proportional gain K_{ip}	1.5332
Integral gain K_{ii}	6250.9

by dividing v_{conv}^* by the input voltage of the power converter v_{in} . After this process, the reference duty cycle, d^* , is obtained.

Tasks

1. Simulate the circuit of Fig. 3 using the controller of Fig. 4 with the parameters of Table 2. Configure the output voltage source to 400V, and set the reference current to zero, $i_L^* = 0$.
2. Apply a reference current step from 0 to 10A. Plot the transient response. **Comment your results.**
3. After the inductor current is set at 10A, apply a step disturbance on the input voltage of 10V.
 - Show the response of the inductor current.
 - Show the behaviour of the duty cycle.
 - **Comment your results.**
4. Similar to the previous task. Once the inductor current is set at 10A, apply a step voltage disturbance of 10V on the output voltage. Compare and comment on the transient response **with and without the output voltage feedforward**.

Part 3

The external output voltage can be implemented once the internal current controller is working properly. In this part, the reference circuit is again the circuit implemented in Fig. 1, in which the capacitor voltage acts as an output filter.

The output voltage regulator is also based on a PI controller, with output current feedforward. The control diagram is shown in Fig. 5, and its parameters are listed in Table 3.

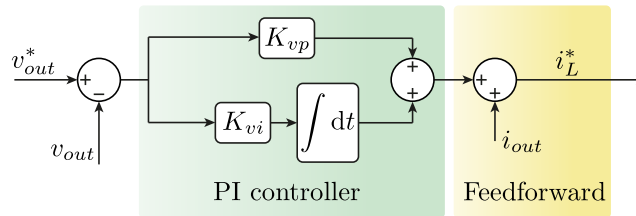


Figure 5: External output voltage (capacitor voltage) regulator with output current feedforward.

Please, observe that the output of the voltage controller corresponds to the inductor current reference. Therefore, interconnect both diagrams properly.

Table 3: Voltage controller parameters

Parameter	Value
Proportional gain K_{vp}	0.0783
Integral gain K_{vi}	30.598

Tasks

In this part, the resistive load will be replaced by a controlled current source in order to emulate a load that consumes power but also regenerates power.

1. Simulate the circuit of [Fig. 1](#), using the cascaded control diagrams of [Fig. 4](#) and [Fig. 5](#). **Do not connect the load**, and apply a voltage reference from 250 to 400V. **Show and comment the transient response.**
2. Regulate the output voltage to 400V, and then connect a load that consumes 50A. Once the transient is achieved, invert the current to -50A. **Plot compare and comment your results,.**